Disclosure Without Inference: LLM Agents Anchor on Disclosed Trade Data When Bargaining

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Abstract

As LLM agents begin to negotiate on behalf of people, the agents on either side may differ far more in capability than human negotiators do, raising the question of which protocols still protect the weaker side [1]. One candidate is an information floor: disclosing data the weaker side could not infer on its own, as US regulators did when publishing corporate bond trades, after which investors' trading costs fell [5]. We test whether such a floor helps an LLM agent. In a calibrated bargaining game, an LLM client buys a Treasury note from a dealer with a private cost; in some conditions it also sees a summary of other dealers' reported purchases (the band). A Bayesian client facing the same dealer gains 11% of the surplus from the band. The LLM client (Ornith-1.5-35B) gained nothing: its share changed by −7.3 percentage points (95% CI [−19.0, 3.8]), in a pre-registered test that held on unseen scenarios. The client did respond to the band, but by bidding its numbers rather than reasoning with them: a quarter of its counteroffers sat on the band's lowest price, where most were rejected. Its beliefs moved half as far as a Bayesian's and only weakly guided its bids. Disclosure cannot substitute for inference the agent does not perform. Marketplaces that admit negotiating agents may therefore need to regulate the agents themselves, for example by requiring them to pass capability tests, such as games with a Bayesian benchmark like ours, before they trade for others.

1 Introduction

The experiment is a dyadic negotiation, where a client agent buys an off-the-run US Treasury note from a single dealer with a private cost. In some conditions, it also sees a summary of other dealers' reported purchases. We build a calibrated bargaining game with a Bayesian benchmark, under which the disclosure raises the client's share by 11% of the surplus on our test scenarios. We then test a 35B open-weight model that crosses disclosure with belief elicitation.

Disclosure did not reliably improve the LLM's outcome (−7.3 percentage points, 95% CI [−19.0, 3.8]), well short of the Bayesian gain. The client did respond to the disclosure, but by bidding the numbers it was given: its counteroffers moved towards the disclosed typical cost and often sat on the bottom of the disclosed range, where most were rejected. Its stated beliefs did little to drive its actions. In conclusion, an information floor cannot substitute for inference the agent does not perform: disclosure protects a weaker agent only if it can use what is disclosed, so protocols for agent negotiation may also need to bound the capability gap between the agents themselves.

2 Background

Bargaining with private values. When both sides of a trade hold private values, no bargaining procedure can guarantee that every worthwhile trade happens while giving both sides an incentive to be truthful [9]. How the surplus is split then depends on the protocol and on what each side knows. Information design studies how a third party can change outcomes by choosing what to reveal [7]. In corporate bonds, the introduction of trade reporting through TRACE lowered investors' trading costs [3, 4]. Those studies measure market-wide averages, whereas we look inside a single negotiation.

LLM agents in bargaining. Lewis et al. [8] trained negotiation agents end to end. Qian et al. [10] compared humans, LLMs and Bayesian agents in a multi-player trading game and found that similar surplus can hide different strategies: LLMs made concessionary offers that were usually accepted. Like them, we use induced values and a computed benchmark, but we hold the agent fixed and vary its information.

Bayesian theory of mind and anchoring. Inferring another agent's private state from its actions can be modelled as Bayesian inverse planning [2, 12]. A client able to read a dealer's cost from its quote would need no disclosure. People, by contrast, often anchor on salient numbers [11], and LLMs show similar biases [6]. Our design distinguishes Bayesian use of a disclosure from anchoring on it.

3 Bargaining Game with a Bayesian Benchmark

In this game, a client buys \$10M of an off-the-run 10-year note from one dealer in a game limited to three moves (Figure 1). First, the dealer quotes a price p. The client either accepts p, or makes a counter-offer $q < p$. The dealer accepts q if and only if $q \geq c$, its private cost.

The client has a private limit r, the most it will pay, and earns r minus the price, or zero if there is no trade. Both sides see an indicative evaluated price m. Our primary outcome is the client's share of the surplus, zero if there is no trade, or its capture as defined as follows:

$$\text{Capture} = \frac{r - \text{price}}{r - c}. \tag{1}$$

Calibration. Each scenario draws the dealer's cost c below m and the client's limit r above it, each at a random distance (LogNormal, median $1g$). Both sides know these distributions. The unit g, about \$3,400 on \$10mm, is one tenth of the note's daily price volatility, which puts it within the range of real dealer margins.

Disclosure. Ten other dealers' reported purchases are centred on the market-wide typical cost, from which this dealer deviates by a

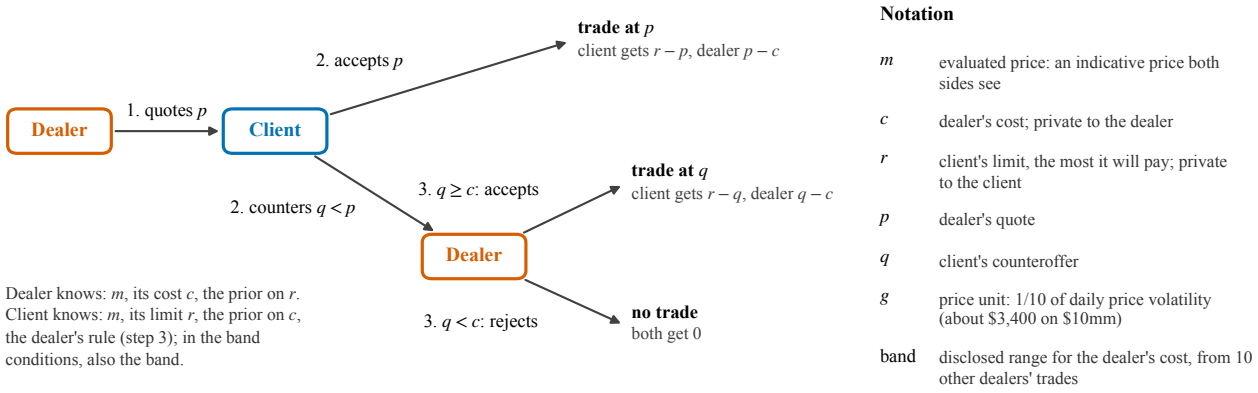


Figure 1: The bargaining game. The dealer quotes a price p; the client accepts it or counters with a lower price q; the dealer accepts the counter only if it covers its cost c, and its decision is final. A trade at price x pays the client $r - x$ and the dealer $x - c$; no trade pays both zero. Each side knows its own private value (c or r) and only the distribution of the other's. In the band conditions the client also sees the band.

random amount $d \sim \mathcal{N}\left(0, (0.25g)^2\right)$. In the simulations where the band was disclosed, the client reads:

> "Reported purchases by other dealers of similar off-the-run notes yesterday (10 trades) averaged X below their evaluated prices. Applied to today's evaluated price, that implies a typical dealer cost of avg. Individual dealers' costs vary around the typical level, so this dealer's own cost is within [low, $high$] about two times in three."

The band's centre is an unbiased but noisy reading of this dealer's cost, with standard deviation $s = 0.356g$, against a prior standard deviation of $0.604g$. The stated range, $avg \pm s$, contains the true cost 68% of the time, so the statement is exactly true.

Rational Client. As a benchmark, we compute the play of a Bayesian client. Suppose the client holds a belief $G(q)$, the probability that the dealer's cost is at or below q. Then its best counter q^* is the counter that maximises $(r - q)\,G(q)$, and the highest quote it should accept is $\theta = r - \max_q (r - q)\,G(q)$. It accepts the quote p if $p \leq \theta$, and otherwise counters at q^*: a lower counter means a better price but a higher chance of rejection. Without the band, G is the prior. With the band, G is the Bayesian posterior.

The Dealer. The same fixed algorithm in every condition. It chooses the quote that maximises its expected profit over the prior on the client's limit, assuming an uninformed client. It rounds the quote to 0.001 points, and accepts any counter at or above its cost. It quotes $m - 0.425g$ in 88% of scenarios, so its quote reveals little about its cost: the quote explains 18% of the variation in the cost.

Value of disclosure. We play the rational client against this dealer at the true cost and limit. Over 1,000 scenarios, the band raises the client's surplus by 11.35% of the total surplus. On our 50 test scenarios it raises capture from 0.621 to 0.732 (+0.111). This is the most a client that uses the band perfectly can gain. It is each side's best response to its stated beliefs, not an equilibrium.

4 Experiments

4.1 Design

Choosing the client. Before the experiment, we chose the client model with a short pilot on separate scenarios. The pilot checked basic competence, for example, that at least 90% of the model's counteroffers were below the quote and within its own limit. Our first choice, Qwen3.6-35B, failed this check (67%). Ornith-1.5-35B passed (92%) and was used throughout. It runs at temperature 0.5 with reasoning off, so it must bargain without step-by-step working.

The Experiment. We varied two things: whether the client sees the band, and whether we elicit its beliefs. Belief elicitation asks for the client's probability that the dealer's cost lies below each of nine prices, once before it sees the quote and once after, so that we can compare its beliefs with a Bayesian client's. From the nine answers we read off the client's best guess of the cost: the price at which it thinks the cost is equally likely to be above or below (the median of its stated belief). Each condition was played on the same 50 scenarios, twice each. The main test (the band raises the client's capture without belief elicitation) and its decision rule were pre-registered before any negotiation with the band. All confidence intervals are 95% bootstrap intervals over scenarios (10,000 resamples).

4.2 LLM Agent Bargaining Results without Disclosure

Without the band, the client almost always made a counteroffer (98–99% of negotiations), while a rational client would usually accept the quote. Its counteroffers were also low: on average $1.3g$ below the evaluated price. About half were rejected, so the client kept only 36–41% of the surplus. A rational client keeps 62%.

The client's counteroffers sit on numbers it has been given. Without belief elicitation, 77 of its 98 counteroffers land on one of two

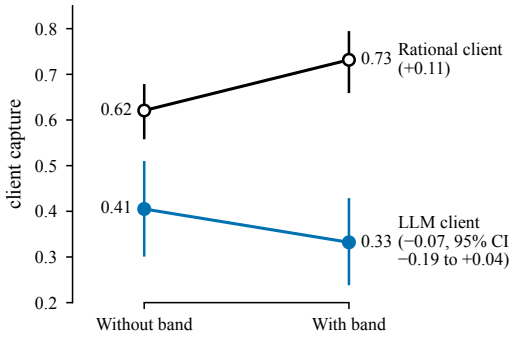


Figure 2: Client capture without and with the band. Rational client: open markers; LLM client, without belief elicitation: filled markers. The band raises the rational client's capture by 11% of the surplus; the LLM's change is not significant.

prices from the briefing: the typical dealer cost, 1g below the evaluated price (about $3,400 on the $10mm trade; 46 counteroffers), or the low end of the briefing's 80% range of costs, 1.9g below (31 counteroffers; Figure 3a).

Belief elicitation itself also supplied prices to bid. It asks for the probability that the dealer's cost is below each of nine listed prices, and with elicitation, half of the client's counteroffers sat exactly on one of them (1.25g, 1.5g or 2g below the evaluated price). Without elicitation, the client almost never bid these prices.

Its stated beliefs do not drive its bids. Its beliefs imply a counteroffer about 0.47g below the evaluated price, but they do not predict its actual counteroffers (rank correlation 0.08). Nor do its counteroffers rise with its own limit.

4.3 Evaluating if an Information Floor Helps

The band did not improve the LLM client's outcome. Without belief elicitation, the client kept 41% of the surplus without the band and 33% with it (Figure 2). The difference, −7.3 percentage points (95% CI [−19.0, 3.8]), is not significant, so our pre-registered main test is not supported. A rational client facing the same dealer raises its share from 62% to 73% with the band, a gain of 11 percentage points. That gain lies above the LLM's entire confidence interval. So, even at best, the LLM gained about a third of what the band offers. With belief elicitation, the band made no difference either (−0.2 points, 95% CI [−10.2, 9.7]).

4.4 How the LLM Agent Used the Band

The LLM agent reacted to the band, but by bidding its numbers rather than reasoning with them. As predicted, its counteroffers moved towards the band's centre, by 0.16g without belief elicitation (95% CI [0.08, 0.24]) and 0.26g with it ([0.19, 0.33]).

An exploratory look in Figure 3 shows this. Without belief elicitation, 26% of counteroffers sat exactly on the bottom of the band's stated range, against 1% without the band. The dealer's cost lies below that point only about 16% of the time, so 77% of these counteroffers were rejected, which explains why the estimated effect of

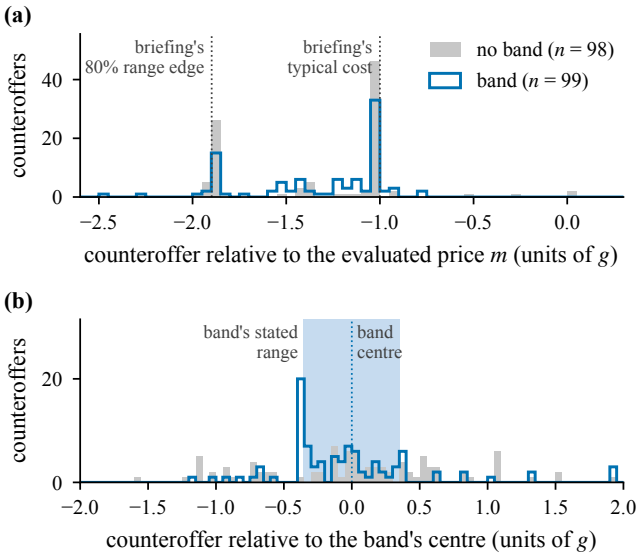


Figure 3: Where the client bids, without belief elicitation. (a) Without the band (grey), counteroffers pile up on numbers in the briefing. (b) With the band (blue), they move onto the band's own numbers, most often the bottom of its stated range (shaded).

the band is negative. The band did not replace the briefing's numbers: the typical cost from the briefing was still the client's most common counteroffer. And, as without the band, its counteroffers did not depend on its own limit, although a rational client with a higher limit bids higher to make a rejection less likely.

4.5 What the LLM Agent Believed

Belief elicitation shows that the band shifted the client's beliefs but did not make them more accurate. With the band, the client's best guess of the dealer's cost (the median of its stated belief) moved 0.23g towards the band's centre (95% CI [0.14, 0.34]), about half as far as a Bayesian client's (0.46g; Figure 4). Yet its belief as a whole was no closer to the Bayesian belief (change in mean squared error +0.012, 95% CI [−0.004, 0.030]). Its counteroffers, meanwhile, moved 0.25g closer to a rational client's (95% CI [0.15, 0.33]). Behaviour that moves towards the rational choice while beliefs stay no more accurate is the pattern suggests anchoring, not Bayesian use. Its bids also did not follow its own beliefs: without the band, the counteroffer a rational client holding the client's stated beliefs would make did not predict the one it made (rank correlation 0.08).

The client also weighed evidence the wrong way round (Figure 4). A Bayesian moves its best guess in proportion to how much a piece of evidence reveals. The dealer's quote reveals little, because the dealer quotes the same price in most scenarios whatever its cost: even a client that knows the dealer's quoting rule moves its guess only 0.08g. Yet after seeing the quote, the client roughly halved its probability that the cost was at least 1g below the evaluated price

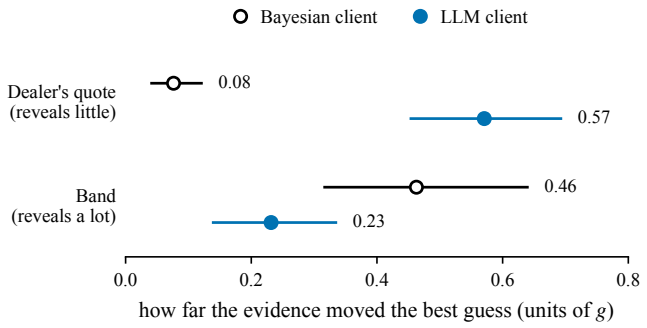


Figure 4: How far each piece of evidence moved the client's best guess of the dealer's cost (the median of its stated belief), in the conditions with belief elicitation. The dealer's quote reveals little about its cost: a Bayesian client that knows the dealer's quoting rule moves its guess only $0.08g$, but the LLM moves it $0.57g$. The band reveals much more: the Bayesian moves $0.46g$ towards its centre, the LLM only $0.23g$. Means over 50 scenarios with 95% CIs.

(from 0.71 to 0.36), moving its best guess $0.57g$. The band reveals far more, yet moved its guess only $0.23g$, half as far as a Bayesian's.

4.6 Confirmation from Unseen Scenarios

We repeated the experiment once on 20 scenarios never used during development, with the analysis code unchanged. The main findings held (Figure 5). The band again had no reliable effect on the client's share of the surplus (panel a): its confidence interval spans zero in both sets. The client again moved its bids towards the band's centre (panel b), confirming that it anchors on the numbers it is given. And its beliefs again lagged the Bayesian client's (panel c): the band moved its best guess half as far as a Bayesian's in the main experiment, and not at all on the new scenarios.

5 Limitations and Future Work

Limitations. The game isolates the effect of information at the cost of realism. There is a single dealer, so the client has no competing quotes, its main protection in real markets; it makes one counteroffer, where real negotiations can go back and forth; and walking away is worth nothing, where a real client could trade elsewhere. The dealer is a fixed algorithm, not an agent: it neither adapts to nor exploits the client, and its quote reveals little about its cost, so we can detect a client over-reading the quote but not reading it skillfully. Costs, limits and reported trades are synthetic; only the price scale is calibrated to market data. The Bayesian benchmark is a best response to stated beliefs, not an equilibrium. Finally, we test one model, with reasoning off, under one prompt, so our results describe this model in this mode.

Future work. Future work should test whether the failure is specific to this setup. First, every client is told the distribution of dealers' costs, as parity with the Bayesian client requires, but the briefing states it as concrete prices, and these became the client's most common bids; stating the same distribution without prices would show

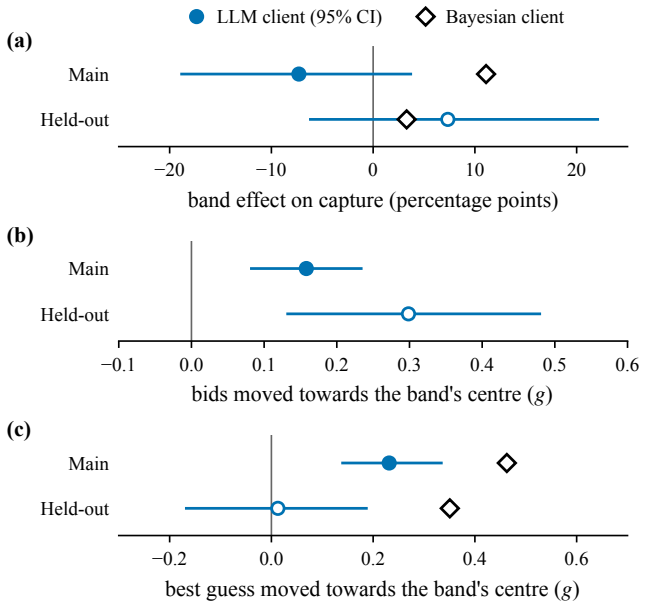


Figure 5: Main experiment (filled; 50 scenarios, two runs) against 20 held-out scenarios (open; one run), with 95% CIs. Diamonds: Bayesian client. (a) The band's effect on the client's share of the surplus. (b) How far bids moved towards the band's centre. (c) How far the client's best guess of the cost moved towards the band's centre.

whether the anchoring depends on being given a price. Second, repeating the experiment with reasoning on would show whether step-by-step working lets the client use the band. Third, testing other model families and sizes would show whether any LLM agent can use an information floor to close a capability gap.

6 Conclusion

Disclosing other dealers' trade data did not reliably improve an LLM client's outcome, and it delivered significantly less than a Bayesian client gains from the same information. The client did not ignore the disclosure; it treated the disclosed numbers as prices to bid. Its failure is one of inference. It over-read the dealer's quote, which says little about the cost, and under-used the band, which says much more: the reverse of how a Bayesian weighs evidence. For this model, then, the floor supplied information but not the inference needed to use it.

Two lessons follow for protocol design. First, how information is presented matters as much as whether it is presented: an agent that anchors will bid whatever number it is shown, so a disclosure stated as a single price may do more harm than one stated as a range or a distribution. Second, asking an agent about its beliefs is not neutral: the prices in our elicitation questions themselves became anchors. More broadly, if agents cannot use what is disclosed to them, information floors alone cannot protect the weaker side, and protocols for agent negotiation may also need to bound the capability gap itself.

References

[1] AGI Institutions Wiki. 2026. Agent Negotiation & Bargaining. Draft page; no author or publication date shown. Retrieved November 5, 2026 from https://www.agi-institutions.org/cell/dyadic/preferences

[2] Chris L. Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B. Tenenbaum. 2017. Rational Quantitative Attribution of Beliefs, Desires and Percepts in Human Mentalizing. *Nature Human Behaviour* 1, 4, Article 0064 (2017). doi:10.1038/s41562-017-0064

[3] Hendrik Bessembinder, William Maxwell, and Kumar Venkataraman. 2006. Market Transparency, Liquidity Externalities, and Institutional Trading Costs in Corporate Bonds. *Journal of Financial Economics* 82, 2 (2006), 251–288. doi:10.1016/j.jfineco.2005.10.002

[4] Amy K. Edwards, Lawrence E. Harris, and Michael S. Piwowar. 2007. Corporate Bond Market Transaction Costs and Transparency. *The Journal of Finance* 62, 3 (2007), 1421–1451. doi:10.1111/j.1540-6261.2007.01240.x

[5] Michael A. Goldstein, Edith S. Hotchkiss, and Erik R. Sirri. 2007. Transparency and Liquidity: A Controlled Experiment on Corporate Bonds. *The Review of Financial Studies* 20, 2 (2007), 235–273. doi:10.1093/rfs/hhl020

[6] Erik Jones and Jacob Steinhardt. 2022. Capturing Failures of Large Language Models via Human Cognitive Biases. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*. Curran Associates, Inc., 11785–11799. https://proceedings.neurips.cc/paper_files/paper/2022/hash/4d13b2d99519c5415661dad44ab7edcd-Abstract-Conference.html

[7] Emir Kamenica and Matthew Gentzkow. 2011. Bayesian Persuasion. *American Economic Review* 101, 6 (2011), 2590–2615. doi:10.1257/aer.101.6.2590

[8] Mike Lewis, Denis Yarats, Yann N. Dauphin, Devi Parikh, and Dhruv Batra. 2017. Deal or No Deal? End-to-End Learning of Negotiation Dialogues. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Copenhagen, Denmark, 2443–2453. doi:10.18653/v1/D17-1259

[9] Roger B. Myerson and Mark A. Satterthwaite. 1983. Efficient Mechanisms for Bilateral Trading. *Journal of Economic Theory* 29, 2 (1983), 265–281. doi:10.1016/0022-0531(83)90048-0

[10] Crystal Qian, Kehang Zhu, John J. Horton, Benjamin S. Manning, Vivian Tsai, James Wexler, and Nithum Thain. 2026. Strategic Tradeoffs Between Humans and AI in Multi-Agent Bargaining. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26)*. Association for Computing Machinery, 1625–1646. doi:10.1145/3742413.3789078

[11] Amos Tversky and Daniel Kahneman. 1974. Judgment under Uncertainty: Heuristics and Biases. *Science* 185, 4157 (1974), 1124–1131. doi:10.1126/science.185.4157.1124

[12] Tan Zhi-Xuan, Jordyn L. Mann, Tom Silver, Joshua B. Tenenbaum, and Vikash K. Mansinghka. 2020. Online Bayesian Goal Inference for Boundedly Rational Planning Agents. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. Curran Associates, Inc., 19238–19250. https://proceedings.neurips.cc/paper/2020/hash/df3aebc649f9e3b674eeb790a4da224e-Abstract.html